

Department of Computer and Information Sciences CS958 – Project

What is required for each dissertation type

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1 Introduction

This document provides more information about the content required for each dissertation type. Please refer to the document on [Dissertation Types](#) to determine the appropriate type and marking scheme for your dissertation.

2 Type 1: An extensive literature survey

All dissertations require a literature review which provides evidence of independent reading. This informs you as a researcher and forms the theoretical basis for your work. It also guides the direction of your research and may provide secondary data, e.g. for discourse analyses. The extended literature review requires greater depth of material and greater consideration of the material. It is not simply a summary of available literature but should present a novel interpretation of the literature based on your insight and reading.

Key questions to ask in an extended literature review are:

- What is already known about this topic?
- What concepts and theories are important?
- What research methods and approaches have been adopted?
- Are there any significant controversies?
- Are there any inconsistencies in findings?
- Are there any unanswered research questions?

Acquiring the literature

Most searches are unsophisticated and tend to be recall oriented, trying to obtain all the literature rather than focusing on the most useful literature on a topic. A second weakness in extended literature reviews can be simply collecting literature and then trying to 'force' a narrative on the literature collected. Do not just rely on Google and other search engines. Do look at specialist tools such as Google Scholar and use [subject specific databases](#) (<https://www.strath.ac.uk/professionalservices/library/researchrevise/collections/eresourcesonlineresources/> and <http://guides.lib.strath.ac.uk/cis>).

Organising the literature

You need to develop a clear and compelling framework around which your analysis is structured. The simplest approach is to use key themes related to the research questions. A chronological approach may also be useful in some cases to show how an

area has developed over time. Initial literature searches may focus on gathering review material and other documents that help orientate you in the topic before you start reading in detail.

Do keep notes on the material you read as extended literature reviews need to show the source material has been critiqued. Questions to ask are:

- What are the strengths and deficiencies of the material you are reading?
- Are there any errors or significant limitations?
- What are the relationships with other work?
- What are the levels of consensus and coherence within the papers you are reading – are the papers tackling similar, identical, or completely different research questions?

Do also keep records on what sources you used and how you obtained these. This record is important in writing your methodology section.

- What sources did you use –databases, libraries, agencies, websites, informal sources, grey literature?
- Why did you use these sources?
- What search strategies did you use?
- What criteria did you apply for inclusion and exclusion of material?
- How useful, reliable, etc., were the sources?

Types of literature review

There are two major types of extended literature review: **narrative** and **systematic**.

Narrative literature reviews are interpretative and synthetic in that they describe a process of discovery to build a picture of an area of knowledge based on detailed reading and understanding of appropriate literature. They lead to better understanding of a field of study rather than necessarily providing clear answers to research questions. They can be seen as a form of meta-ethnography in that they describe how researchers have viewed, tackled, and answered research questions as expressed through the research literature. Such reviews are common in social sciences and qualitative research.

A common critique aimed at such reviews is that they are opportunistic and selective, that is researchers pick and choose which literature fits their existing perceptions and therefore the reviews are open to bias on the part of the researcher. Any review must be objective and deal with the literature as it appears, noting contradictions in the literature and honestly reflecting the current state of knowledge rather than trying to

force an opinion on the literature. However, personal interpretation, the form of new insights, is very valuable and characterizes a good literature review.

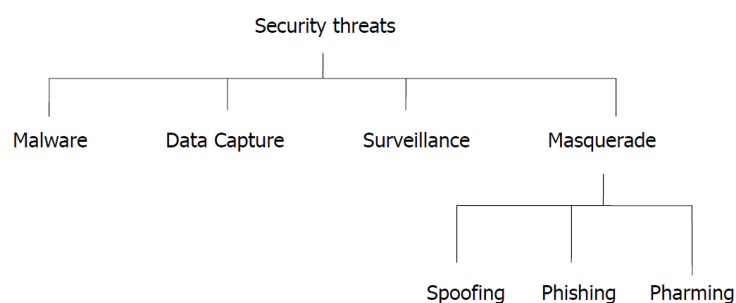
Critiquing literature is very important. Even though work is published in peer-reviewed form does not mean it is correct or valuable. However, we should be fair when critiquing others' research; knowledge is always in a state of flux and earlier researchers did not have the benefit of later researchers' insights and knowledge.

Systematic literature reviews are usually positivist in approach: trying to build patterns based on evidence and are, hence, meta-analytic. They are more common in natural sciences to provide a scientific and transparent process based on summarizing (usually large) numbers of studies on a common theme or research question. Typically, systematic literature reviews lead to a specific addition to knowledge in a field and are more suited to answering specific research questions rather than general ones, e.g. which approaches search system evaluation are best, do men and women use technology differently, what factors influence use of public libraries, and are useful for forming evidence-based conclusions to such questions.

A common critique of this type of review is that they often claim to be comprehensive, but the parameters are chosen by the researcher, e.g. reviewing children's success in using search systems needs a definition of what constitutes being a child and what counts as a search system, both decisions are made by the reviewer and different choices may lead to different results.

Analysis

There are different ways to analyse the literature and present your findings to the reader. One approach is to provide a **taxonomic** description (as shown below) where high-level concepts, such as security threats, are broken down into sub-concepts, such as malware, which can be described in separate sections. Such an approach can be useful in leading the reader from general concepts through to more specific concepts or implementations.



A second approach is a **matrix** approach where concepts, such as freedom of information, data protection and intellectual property, are compared against major variables, such as implementation in different countries. This leads to a compare

approach to the literature, highlighting major differences between the variables for each concept.

National Approaches to Information Legislation

	US	UK	China
Freedom of Information			
Data Protection			
Intellectual Property			

A second example is shown below where different media are compared against implementations of search systems to help retrieve these media.

Information Retrieval from Different Media Types

	Statistical	Probabilistic	Semantic
Text			
Images			
Sound			

A third approach, suitable in particular for comparing studies on a common theme, is the template approach. A **template** describes major variables, such as sample size, relevant to each study and each template provides data on a single study. This approach can also be useful for comparing between studies, identifying missing data in studies and comparing the values for each variable. The templates are a way of organising notes on the studies; the dissertation should still provide a textual analysis of the findings from the templates.

Analysis – Template Approach



Citation	
Sector	
Country	
Sample Size	
Instrument	
Response Rate	
Perceived Advantages	
Perceived Disadvantages	

Documents as evidence

In the literature review the focus is on what people have said about the topic but documents (text and non-text) may be the topic of study themselves. For example, the review may be on certain types of documents such as legislation, official document, business documents, emails, blogs or tweets and may concentrate on physical characteristics (e.g. publishing and printing history), digital characteristics (e.g. properties), or content (e.g. how it is said, and what is said). Content itself may focus or consider aspects such as register (context and purpose), voice (passive/active, aggressive/defensive), interaction between document sources (conflict, consensus), rhetoric (metaphor, alliteration), vocabulary (orientation, evolution), consistency (intra- and inter- document), signatures (authorship, tag clouds) or semiotics (symbols, designs).

Summary

Extended literature reviews involve the interpretation of a larger body of literature. The outputs should not just be a summary of what others have said but should provide new insights and perspectives. This may involve content analysis, which can also be applied in surveys and case studies

3 Type 2: Case study based

A case is an event or set of events which occur(s) within a specific context, for example:

- Case handling, to deal with claims in an insurance company
- Medical case management, to ensure that there is appropriate diagnosis, treatment, and care
- A criminal court case, to bring an accused to, and through, trial

Case-studies are complex and intricate studies which are often location and/or organisation based. They are time-sensitive in that they investigate current, rather than historical, situations which involve a combination of people, processes, and behaviours. They may involve quantitative data but typically involve a lot of qualitative data. Case studies are appropriate for in-depth, rich investigations of complex phenomena and this often means investigating fewer samples (cases). However, the small number of cases does not mean the case study is simpler or easier than other types of dissertations.

Here is an example of a **research context** for a case study dissertation

Many enterprises are considering hot-desking as a means of reducing the costs of office space which often lies unused. Technology is seen as key contributor in delivering more flexible working, including home-working and off-site working.

This leads into a **research problem**:

The introduction of hot-desking requires employees to alter working practices and introduces new issues ranging from information security, through staff appraisal, to capacity planning

And subsequent **research questions**:

- To what extent have expected business benefits (as specified in the business case) materialised, and why?
- How do employees view the shift to hot-desking?
- How might the transition to hot-desking be improved?

A literature review on this problem, as with other case studies, needs to consider different viewpoints, including hot-desking literature, remote working literature, organisational change literature, and change management literature. It may also include literature from professional bodies such as the Chartered Institute of Personnel

and Development and research institutes such as the Institute of Work Psychology (Sheffield) who as produce official reports on topics such this.

Ethnographic approaches

A common data gathering used in case studies is ethnography. Ethnographic approaches use observation and associated field notes, interviews and associated transcripts, internal documents (business cases, policy statements, employee satisfaction surveys, etc.). Time constraints may lead to more focused studies (micro-ethnography), e.g. how do staff in call centres deal with work problems in an environment of constant interruptions and monitoring?

Observations may be closed (where we are only interested in certain types of behaviour that we know are important) on open (where we observe all behaviour). Observations may also be structured we observe certain groups at pre-determined times or unstructured where we place no restrictions on the observations. Observers may be participants in the observation scenario, e.g. acting a librarian in a library observation, or non-participant where the role is purely observational. The observations may also be overt (where observation is occurring) or covert (where the observation is hidden from those being observed). These choices must be carefully considered and justified both ethically and in terms of the research being conducted. Consider the four scenarios below:

	Open	Closed
Overt	Queuing Behaviour in a Station	Employee Attitudes to Hot-Desking
Covert	Reactions to Self Service in a Library	Racism amongst Delegates at a Political Conference

In the first situation there is no need to be covert in terms of observation as this will not change behaviour. In general, overt observation is more ethically preferable to covert observation so should be chosen unless there is a good reason to be covert. In this situation there might not be any clear idea of what behaviours may arise in such a situation, so open observation is appropriate. In the fourth situation (politics) closed observation is appropriate but also covert observation as attitudes such as these may not be openly displayed in the presence of an outsider.

Situations that are appropriate for open coding are ones where there may be fewer pre-determined behaviours of interest whereas the ones appropriate for closed coding are those where the data of interest (such as attitudes) can be guessed in advanced.

In any ethnographic study we have to ask what types of observation are appropriate, what are practicable (in terms of time, researcher effort and data to be gathered), what is the influence of the participant on the situation (might your presence change the situation begin observed and therefore misrepresent reality), and also the influence of the situation on you (some situations are potentially distressing and should be approached with care).

Interviews, surveys, and other data gathering techniques may also be employed within case studies.

Data analysis

An in-depth description of a particular case study using qualitative analysis may be used to test a well formulated theory, compare a situation over time, look at extremes, look at completely new conditions, compare against previous case studies or describe typical situations. Comparative analysis of two, or possibly three, case studies may also be undertaken in different sectors, cultures, or jurisdictions. Analysis of more than three situations moves into the realm of surveys which will be considered later in this guide.

The qualitative techniques used in analysing documents are all applicable: template, matrix, taxonomic or content analysis, but other coding may also be required such as body language, facial expressions, event types, etc.

Case studies may also consider factors such as:

- Finances –identifiable savings; set-up costs; employee savings; insurance
- Technology –security; functionality; licensing; fixed versus moveable devices; telephones
- Information –storage; access; security; privacy
- Operations -capacity planning; bookings; environment; office furniture; hygiene; human factors; personal storage; replacements
- Human resources –supervision; appraisal; health and safety; education and induction; consultation; job descriptions; special needs
- Culture –team building; knowledge transfer; (im)permanence; values; pervasiveness; shared versus personal space

Recommendations

Recommendations can be of several types, e.g.

- Testing, e.g. the extent to which a business theory reflects reality

- Explanatory: an understanding of how consensus, conflict, co-operation, etc., operate in complex situations
- Learning: how to deal with similar situations in the future

Summary

Case studies are complex, detailed investigations which may require several methods to be employed. Much of what is analysed is documentary (transcripts, field notes, business documents) and may include observations. Case studies lead to insights and learning rather than general laws but are very valuable in creating understandings of social and business phenomena.

4 Type 3: Application based

Applications are written applications, i.e. designed and coded by the researcher. They may make use of existing software components but should have some clear element of original software development. These may be desktop applications, mobile applications, or web-based applications.

All applications, like other dissertations, have a **research context**, for example

A pressing challenge for museums is how to present museum objects in an intriguing way to engage visitors. With added pressure to compete with the entertainment industry, museums are adapting to incorporate interactive technologies in the museum space. A particular challenge is developing smart museum objects: objects which have digital content embedded and can interact with visitors. These can enhance the visitor experience in numerous ways helping the visitor better understand the object.

which in turn leads to a **research problem**:

This project will develop smart objects for the museum sector. It will, through developing smart objects, investigate what styles of engagement are appropriate for the museum sector and how such objects should be evaluated.

and **research questions**:

- What styles of engagement are appropriate for museum smart objects?
- What technology is suitable is appropriate for such objects?
- How can we make such objects sufficiently robust for repeated use within a typical museum setting?
- How can smart objects best be evaluated?

Literature reviews for dissertation can be based on the peer-reviewed research literature but for very recent problems there may be little to guide us. Alternatives are market analyses (what other applications are already available and what is the gap in the market), developer fora such as and product documentation.

An alternative may be also to conduct studies as to learn more about usage requirements such as usability studies or existing studies such as the iPhone-Nielsen Norman Group seminars.

Development models

Each application project needs some form of methodology to guide and document the process of developing a solution. Typical process models include the standard waterfall models, evolutionary prototyping, agile development, etc. The process model should be described and motivated.

Within any process model there are four basic stages: design, build, test, and deploy.

Design

The design phase of an application dissertation should provide a description of the requirements specification and modelling based on the literature review, going from high-level conceptual design to lower-level design diagrams, using a standard notation such as UML. Supporting material, such as class diagrams, can be relegated to Appendices.

The design material should also describe technical requirements, e.g. particular operating systems, necessary for the application.

Build

The build, or implementation, phase should provide descriptions of the development tools and languages used to implement the solution in a level of detail that allows the reader to reproduce the application. It is not necessary to describe each line of code, but you should describe in the main body of the report any code that is particularly interesting or challenging to develop. All code should be submitted together with your dissertation at submission time.

Test

The test component should describe unit and systems testing plans and results; user acceptance testing document and results and, where appropriate, performance testing results. Detailed test plans can be presented in appendices with summaries of the results presented in the main body of the report.

Deploy

The final stage in any application is deployment either through a formal route such as AppStore or to a user. This is not necessary for your dissertation, but your dissertation should describe how a user can install and run your application, including a user guide in the Appendices.

Evaluation

The test component should describe the testing strategies you used to test the correctness of your system, but it is essential to also conduct evaluations on the application to test other characteristics of your solution. There are a range of evaluation approaches including: usability studies and user opinion studies.

Usability studies are systematic observation under controlled conditions where we test how easy an application is to use and to identify common mistakes users may make when operating a system. It is important to remember that it is the systems being tested

rather than the users. Good usability tests should involve a sample of the intended user group, but individuals should not be used in multiple tests to avoid learning effects.

Usability studies should provide some form of usage scenario to allow users to use the system in a natural way, e.g. use this mobile phone to phone your best friend. The purpose of the application *may* be provided to the user although do remember that many applications do not come with instructions and users often do not read help instructions.

Studies should have some metrics, e.g. time taken to complete tasks, accuracy, or number of mistakes, that will help decide whether the application is a success. We can gather data through several techniques such as log analysis, think-aloud, video, observations, or interviews.

User opinion studies gather information on an application. They are like surveys in that they gather subjective data but here they gather data on an application that participants are asked to use. In opinion studies we may gather information on functionality, utility, design, aesthetics, emotional response, intended use, speed or how likely participants would be to use/buy the application.

Summary

State-of-the-art technologies may not yet be reflected in academic literature and this dissertation type involves the creation of a new application. In the dissertation it is important that you specify the methods you have used to develop the application, and why you think they are appropriate. Testing and evaluation are also methods and need to be described and justified. As with other dissertation types it is only the dissertation that is assessed so the text is particularly important in conveying your achievements.

5 Type 4: Evidence based

These projects collect and analyse data to answer their research questions. They may take the form of interviews, traditional questionnaire-based data gathering tools, focus groups or analysis of existing datasets. They cross-sectional studies where researchers are interested in variation across multiple cases. These studies are based on data collected at a particular point in time for each case. Longitudinal studies are also possible, but rarer, and look at data collected for each case at intervals.

In surveys we are looking mostly at quantifiable data that is based on variables that are non-manipulable, that is variables as they occur naturally rather than manipulating them to see their effect on other variables. The latter approach is the experiment type of dissertation. What all surveys try to achieve is to look for patterns of association in the data collected to draw conclusions about variables relate to each other.

Here is an example of a **research context** for a survey-based approach:

There are several hundred search engines which cater for generalist to specialist information needs. There are also several meta search engines which combine results obtained by different search engines. Despite the availability of choice most users utilise Google as it is often the default search engine on web pages and has global brand recognition.

Which leads to **research problem?**

Google is a search engine of choice, but it is less clear why users ignore other, potentially more effective, engines.

and associated **research questions**

- Which search engines do people use?
- What do they use the engines for?
- Why do they use the engines they do?
- What features do users expect a search engine to offer?

A literature review might consider benchmarking literature, search engine evaluation literature and search strategy literature.

There are many tools that may be used in a survey to gather data such as questionnaires, structured observations, interviews, and content or statistical analyses.

Questionnaires and interviews

Questionnaires may be utilised via the web using tools such as SurveyMonkey, email, telephone, post, or face-face meetings. All questionnaires and interviews should be piloted to ensure questions are understandable and to test timings. They should ask essential questions, i.e. those which provide data you will use and should include clear instructions about how to respond and make it clear that a respondent is not required to answer any questions. The latter is essential for ethical clearance for any questionnaire.

Good practice includes:

- Developing consistency across questions; consistency creates habitability.
- Concentrating primarily on closed questions which are easier to answer and are more likely to be answered.
- Leaving room for people to write if using open questions.
- Trying to have a logical flow of linked questions.
- Aspiring to moral neutrality and balance as 'loaded' or judgemental questions can alienate respondents.
- Thinking carefully about questions which could be construed as overly intrusive.
- Thinking carefully about questions for which more than one response is invited.
- Thinking about the length of the questionnaire as respondents may get fatigued and stop answering.
- Thinking about the ability of respondents to answer the question.
- Thinking about leaving demographic information to the end
- Trying to start with less challenging questions to ease respondents into the questionnaire.
- Close with a courteous thanks

Poor practice includes:

- Asking overly broad questions which are difficult and time-consuming to answer.
- Asking long questions which are time-consuming to read.
- Asking complex questions which are difficult to understand.

- Asking double-barrelled or conjunctive questions which can be difficult to answer, e.g. did you find Google useful and efficient?
- Asking leading questions, e.g. do you find Google useful?
- Asking questions that include negatives as these can be more difficult to understand correctly.
- Asking questions with ambiguous terms
- Using acronyms or specialist terms which might not be familiar to respondents.
- Using imbalanced scales
- Using too many Y/N questions as this can lead to respondent fatigue.
- Using overlapping categories

Sometimes ranking may be a better way of dealing with multiple response questions: so rather than asking 'Which of the following search engines do you use (tick all that apply),' ask 'Rank the following search engines in terms of frequency of use.' This gives you two pieces of information for one question, both of which can be coded.

Summary

Evidence-based approaches work with collected data from people, either in the form of answers to given questions or analysis of data created by people for other tasks, e.g. social media. Framing good questions is a fundamental component of all evidence-based techniques as is testing and re-testing the data collection. Questionnaires, which are often completed at a distance and not in the presence of the researcher, also require thoughtful layout and design. As surveys usually cannot be repeated, they require noticeably clear objectives translated into a clear data gathering tool where the analysis method is known in advance.

6 Type 5: Experimental

Experiments involve variables which are manipulated under controlled settings to investigate how the change in one variable, e.g. different system, age group, tasks, affects other variables, e.g. search success, information literacy, user satisfaction. Experiments rely heavily on quantitative data gathering and analysis and preferably involve hypothesis testing.

Four Key Criteria in Quantitative Research Design

There are four main criteria to consider in the design of any experiment:

1. **Replicability**—can the method be repeated (i.e. is the method described comprehensively and unambiguously)? In general, we should be able to provide a detailed specification of our search strategy, our survey method, our testing methods, etc.
2. **Reliability**—are the results of the study repeatable (i.e. if you apply the method again in the same context will you get the same results)? Here the same results do not mean the same numbers - we cannot use the same people as their belief and knowledge states will have altered – but we should expect the same general results. That is, if our first experiments shows that one system is better than another then repeating the study with another sample from our population should also show the same system is better. If not, then we might have problems with our experiment.
3. **Validity**—do the results produced by the method have integrity (i.e. are the research instruments well designed and appropriate)? In general, yes: we should be able to produce robust and effective research methods which are open to scrutiny. However, good methods may be poorly applied so we may wish to convince our audience, or ourselves, that the findings are “credible.”
4. **Generalisability**—do the results have external validity (i.e. can they be applied to contexts other than that in which they were obtained)? In many cases no: our samples are often non-probabilistic and therefore cannot be claimed to be representative. However, we would hope that the results are “transferable”: they may demonstrate trends, patterns, etc., which can be confirmed in other contexts.

Here is an example of a **research context** for an experiment:

Indexing the contents of a database improves retrieval speed as the entire table does not have to be scanned for matching data. However, performance may be compromised as the indexes must be recalculated every time a record is added or updated. In addition, storage requirements grow as the indexes grow, along with associated backups. Text retrieval increases the problem as redundancy is an essential characteristic of language.

which leads to a **research problem**:

The increase in text-related applications on handheld devices (PDAs, e-book readers, etc.) raises issues about performance and optimisation of applications.

and research questions:

- Which of the following techniques produces the best performance: inverted file, content addressable file store, trigrams, byte substitution?
- How does file size affect performance?

Hypotheses:

- CAFS will be more efficient for databases up to 250MB (~1000 documents), byte substitution for up to 1GB (~4000 documents), while trigrams will be more efficient for larger databases.
- All methods will be more efficient than inverted file for each threshold.

In order that there is a fair test, and we can cleanly associate any performance differences to database design all tests will have: one device per algorithm, all operate on the same model configured with the same preferences, applications, etc. and we will use the same test collection (of sufficient size), only text will be used and this will be in same format. Further, we will use the same queries run in the same order.

This standard set-up allows a fair comparison between the techniques which can be used to test the two hypotheses.

Summary

Experiment-based dissertations are few and are closest to empirical scientific research. However, even though few they are valuable in providing precise, statistically tested conclusions to research questions.